

Sasidhar Varada

Embedded Firmware & IoT Engineer



sasidharvarada



sasidharvarada



varadasasidhar



varadasasidhar03@gmail.com

EXPERIENCE

RESEARCH ASSISTANT | SMART CITY RESEARCH CENTER (SCRC), IIIT HYDERABAD

Sep 2021 – Present | Hyderabad, IN

- Deployed end-to-end IoT systems (firmware to cloud) across four verticals: **Water, Air Quality, Smart Spaces, and Wi-SUN Street Lights.**
- Built a **Digital Twin for Smart Water Network** (ZF) — data ingestion, remote pump/valve Actuation Manager, and a predictive simulation engine.
- Developed and deployed **indoor/outdoor air quality nodes** and conducted **oneM2M interoperability** research for IoT-based smart spaces.
- Presented live demos at **IMC 2024 & 2025, ICASSP 2025** and **SPARQ 2026.**

TEACHING ASSISTANT | IIIT HYDERABAD

2025 – 2026 | Embedded Systems Workshop & RAP-IoT

PROJECTS

RTOS-BASED AIR QUALITY DEVICE | FREERTOS, EMBEDDED C/C++, ESP32

- Developed production-grade FreeRTOS firmware with task scheduling, multi-sensor drivers, and fault-tolerant air quality telemetry.

WI-SUN UTILITY NODE MESH NETWORK | WI-SUN FAN 1.1, RASPBERRY PI

- Configured an end-to-end Wi-SUN mesh (FFN/LFN nodes) with SDS011, AHT10/11 and LDR sensors and remote relay control via a Border Router.

RETROFIT SMART WATER METER | EDGE ML/CV, ONEM2M

- Built a camera retrofit for analog water meters running an on-device ML/CV model to read register values and stream telemetry to oneM2M and ThingSpeak.

HARDWARE TEST JIG & CUSTOM PCB | KICAD, RASPBERRY PI, FUSION 360

- Designed a multi-port interface PCB for automated embedded hardware testing via Raspberry Pi, with a custom 3D-printed enclosure.

FLEXIBLE SMART INSOLE | ARDUINO NANO 33 IoT, VELOSTAT, PYTHON

- Wearable gait analysis with 9 Velostat sensors and a live wireless heatmap.

SELF-ADAPTIVE DIGITAL TWIN DEMO BOARD | RASPBERRY PI, ESP32, PYTHON

- Raspberry Pi Digital Twin with ESP32 actuators for predictive maintenance.

AETHERAGENTS – AGENTIC AI PLATFORM | KAFKA, FASTAPI, LLMs, REACT

- Kafka-native, event-driven LLM pipeline (Planner → Retriever → Writer) distributed across VMs, with consumer-group scaling and YAML lifecycle config.

AQI-BASED ROUTE OPTIMIZATION | PYTHON, GRAPH ALGORITHMS, GIS

- AQI-weighted shortest-path routing on campus using deployed sensor nodes.

QUANTUM GATES FOR BEGINNERS | IBM QUANTUM

- Designed basic quantum gate and circuit demos for beginner learners.

PUBLICATIONS

- **IoT 2026** (Accepted) — Digital Twins for Uncertainty Mitigation in Self-Adaptive Smart City IoT Systems
- **FI-Cloud 2026** — Performance Evaluation of Interoperability and Control Implementation for IoT-Based Smart Spaces
- **ICSA 2024** — Architecting Digital Twin for Smart City Systems: A Case Study

SKILLS

PROGRAMMING

C • Embedded C • C++ • Python • Bash

EMBEDDED & HARDWARE

ESP32 • STM32 • Arduino • Raspberry Pi
• FreeRTOS • PCB Design (Eagle)
• Sensor Integration • Qualcomm QIDK

PROTOCOLS

UART • SPI • I2C • CAN • MQTT • HTTP
• WebSockets • oneM2M • Wi-SUN

SOFTWARE & TOOLS

Linux • Git • Docker • FastAPI • Kafka
• MATLAB • Vibe Coding (AI-assisted)

SPOKEN LANGUAGES

English • Telugu • Hindi

EDUCATION

IIIT HYDERABAD

M.S. BY RESEARCH – ECE | 2024 – PRESENT

LPU

B.TECH – ECE | 2023 | GPA 7.41/10

AWARDS

Wi-SUN Contributor Award

Smart City Challenge, Silicon Labs | 2024

Key Contribution Award

Water Digital Twin, ZF Technologies | 2024

Best Promotion of oneM2M

4th oneM2M Hackathon | Dec 2024

Best Education Exhibit

India Mobile Congress | Oct 2024

Wi-SUN Contributor Award

FAN 1.1 Mesh, Silicon Labs | 2023

User Application Award (Wi-SUN)

KETI Hackathon | Jun 2023

CERTIFICATIONS

- **Ericsson Educate** — 5G, AI-ML, IoT
- **Google Cloud Learn to Earn** — 2022, 23

EXTRA-CURRICULAR

- **ISACQ Club** — quantum event organizer
- **NSS Volunteer** — Aksharamala 2025, 26
- **Marathon Volunteer** — IIITH & NMDC
- **Hackathon Mentor** — KIET Kakinada
- **Member** — Kadamba Mess Council
- **Sports** — Basketball, badminton